www.tntextbooks.in

cuticle on the upper epidermis is thicker than that of lower epidermis. The minute openings found on the epidermis are called ***stomata (singular: stoma).*** Usually, stomata are more in number on the lower epidermis than on the upper epidermis. A stoma is surrounded by a pair of specialised epidermal cells called guard cells. In most dicots and monocots the guard cells are bean-shaped. While in grasses and sedges, the guard cells are dumbbell- shaped. The guard cells contain chloroplasts, whereas the other epidermal cells normally do not have them.

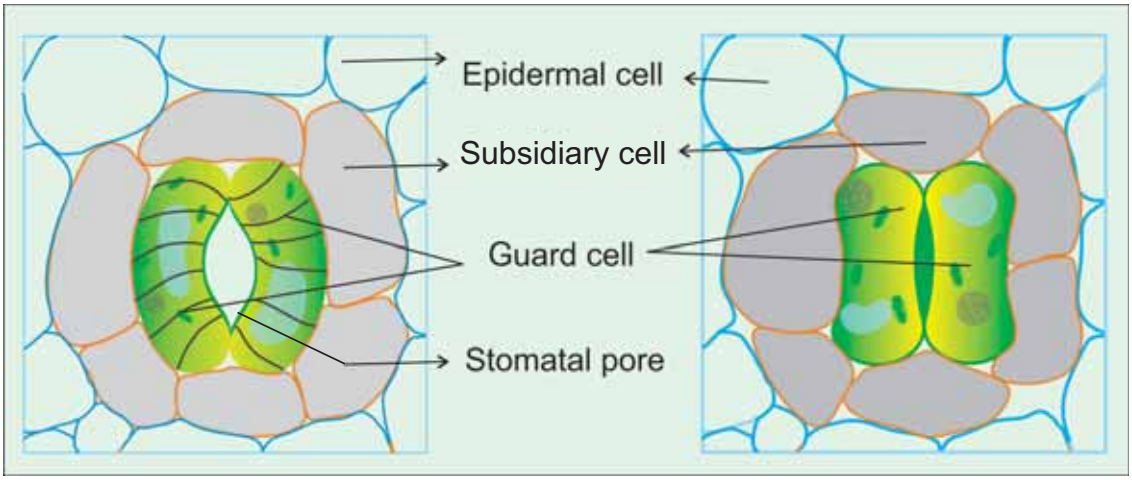



**Figure 9.14:** (a) Stoma with bean-shaped guard cells.
(b) Stoma with dumb-bell shaped guard cells

Some cells of upper epidermis (Example: Grasses) are larger and thin walled. They are called **bulliform cells** or **motor cells**. These cells are helpful for the rolling and unrolling of the leaf according to the weather change. Some of the epidermal cells of the grasses are filled with silica. They are called **silica cells.**

**Check Your Grasp!**

In which group of plants the guard cells are dumb-bell shaped?

***Grasses and sedges***

## Subsidiary Cells

Stomata are minute pores surrounded by two guard cells. The stomata occur mainly in the epidermis of leaves. In some plants addition to guard cells, specialised epidermal cells are present which are distinct from other epidermal cells. They are called **Subsidiary cells**. Based on the number and arrangement of subsidiary cells around the guard cells, the various types of stomata are recognised. The guard cells and subsidiary cells help in opening and closing of stomata during gaseous exchange and transpiration.

## Sunken Stomata

In some Xerophytic plants (Examples: Cycas, Nerium), stomata is sunken beneath the abaxial leaf surface within stomatal crypts. The sunken stomata reduce water loss by transpiration.